

Ten-year-old Chihiro Ogino and her parents Akio and Yūko travel to their new home. Akio, driving down an unexpected road, ends up in front of a tunnel leading to what appears to be an abandoned resort town, which her parents insist on exploring over Chihiro's protests. Upon finding a seemingly empty restaurant stocked with food, Chihiro's parents immediately begin to eat. While exploring further, Chihiro finds an enormous bathhouse and meets a boy named Haku, who warns her to return across the riverbed before sunset. However, spirits begin to appear, and Chihiro discovers that her parents have been transformed into pigs and that she cannot cross the now-flooded river.

Haku finds Chihiro and instructs her to ask for a job from the bathhouse's boiler-man, Kamaji, a yōkai spirit commanding soot sprites known as susuwatari. Kamaji instead asks a worker named Lin to bring Chihiro to Kamaji's master Yubaba, the witch who runs the bathhouse and who cursed Chihiro's parents. Yubaba tries to frighten Chihiro away but eventually gives her a work contract. As Chihiro signs the contract with her name (千尋), Yubaba takes away the second kanji in her name, renaming her Sen (千). She soon forgets her real name, and Haku later explains that Yubaba controls people by taking their names; if Chihiro completely forgets hers like he once did, she will never be able to leave the spirit world.

The other workers, except for Kamaji and Lin, frequently mock Sen. While working, she invites a silent creature named No-Face inside, believing him to be a customer. The spirit of a polluted river arrives as Sen's first customer. After she cleans him, he gives her a magic emetic dumpling as a token of gratitude. Meanwhile, No-Face demands food from the bathhouse workers, granting gold copied from the river spirit in exchange. However, when Sen declines the gold and leaves to find Haku, No-Face angrily swallows some workers.

Sen sees paper shikigami spirits attacking a dragon and recognizes the dragon as a metamorphosed Haku. When the seriously injured Haku crashes into Yubaba's penthouse, Sen follows him upstairs. A shikigami that stowed away on her back shapeshifts into Yubaba's twin sister Zeniba, who turns Yubaba's son, Boh, into a mouse and creates a false copy of him. Zeniba tells Sen that Haku has stolen a magic golden seal from her that carries a deadly curse. Haku strikes the shikigami, causing Zeniba to vanish. Sen and Haku fall into the boiler room, where she feeds him part of the emetic dumpling. He vomits up the seal and a slug that a disgusted Sen kills.

Sen resolves to return the seal and apologize to Zeniba. She confronts an engorged No-Face and feeds him the rest of the dumpling, forcing him to regurgitate the workers. No-Face follows Sen out of the bathhouse, and Lin helps them leave. Sen, No-Face, and Boh travel to see Zeniba using train tickets from Kamaji. Meanwhile, Yubaba nearly orders Sen's parents slaughtered, but Haku reveals Boh is missing and offers to retrieve him if Yubaba releases Sen and her parents. Yubaba agrees, but only if Sen can pass a final test.

The train crosses a sea to a land where Sen meets Zeniba, who reveals that Yubaba used the slug to control Haku. Zeniba tells Sen that she cannot help her parents, but she makes her a magic protective hairband. Using his dragon form, Haku flies Sen and Boh back, while No-Face decides to stay with Zeniba. Mid-flight, Sen recalls falling into the Kohaku River years earlier and being washed safely ashore, correctly guessing Haku's real identity as the spirit of the Kohaku River and restoring his memory.

When they arrive at the bathhouse, Yubaba tests Sen, asking her to identify her parents among a group of pigs. After she answers correctly that none of the pigs are her parents, her contract disappears and she is given back her real name. Haku takes her to the now-dry riverbed and vows to meet her again. Chihiro crosses the riverbed to her restored parents. Shortly before leaving for her new home, Chihiro looks back at the tunnel, still wearing her hairband from Zeniba.